

ViBridge-VQA: Bridging Global and Local Visual Tokens for Parameter-Efficient Vietnamese Visual Question Answering

Nguyen Trung Quoc^[0009-0009-4569-3932],
Nguyen Anh Tuong^[0009-0002-3457-5745], Phan Ba Duc^[0009-0000-1816-9224],
Tran Duc Thinh^[0009-0007-8187-4605], Dang Duy Lan^[0009-0007-8658-6602],
Nguyen Quoc Thinh^[0009-0001-6765-1276], and Tung Le (✉)^[0000-0002-9900-7047]

Faculty of Information Technology, University of Science, Ho Chi Minh City, Vietnam
Vietnam National University, Ho Chi Minh City, Vietnam
{22120071,22120182,22120301,22120347,
22120348,22120412}@student.hcmus.edu.vn, lttung@fit.hcmus.edu.vn

Abstract. Vision-language models (VLMs) such as Vintern-1B have brought strong multimodal abilities to Vietnamese, but adapting them to a new Visual Question Answering (VQA) benchmark remains costly: the usual recipe fine-tunes the language model on many high-resolution image tiles, while task-specific architectures must be trained from scratch. In this paper, we propose **ViBridge-VQA**, a parameter-efficient approach that improves Vintern-1B without updating any of its pre-trained weights. ViBridge-VQA feeds the frozen language decoder two complementary kinds of visual tokens from a single image view: *global* tokens, produced by a small trainable bridge from the [CLS] vector that Vintern-1B discards, and *local* tokens, produced by Vintern-1B’s own pre-aligned projector and pooled to a chosen budget. We treat the number of tokens as an accuracy-cost trade-off and select it as the knee of the Pareto front. On the reasoning-oriented AutoViVQA benchmark, ViBridge-VQA trains only 1.35% of the parameters and passes 50 visual tokens per image, yet it outperforms the officially fine-tuned Vintern-1B, which reads up to 1,792 visual tokens, on seven of eight metrics. Adding a lightweight decoder LoRA raises F1 to 58.2 and gives the highest accuracy, ROUGE-L and METEOR among all compared systems. Our analyses further show that local tokens matter far more than global ones and that they transfer better to unseen Vietnamese VQA datasets than decoder adaptation alone.

Keywords: Vietnamese Visual Question Answering · Vision-Language Models · Parameter-Efficient Learning · Visual Token Budget · Generative VQA

(✉)Corresponding author: Tung Le – lttung@fit.hcmus.edu.vn

1 Introduction

Visual Question Answering (VQA) asks a model to answer a natural-language question about an image, which requires it to perceive the visual content, understand the question and reason over both. The task has progressed rapidly with large vision-language models (VLMs) such as BLIP-2 [11], Flamingo [1] and LLaVA [17], which connect a pre-trained vision encoder to a pre-trained language model and generate free-form answers. Vietnamese has followed this trend: Vintern-1B [6] connects an InternViT-300M encoder [5] to a Qwen2.5-0.5B decoder [25] and is among the strongest open Vietnamese VLMs, while benchmarks such as ViVQA [27], OpenViVQA [20] and the recent reasoning-oriented AutoViVQA [19] make it possible to measure progress in this low-resource setting.

Obtaining a strong model on a new Vietnamese benchmark, however, is still expensive. The common recipe fine-tunes an existing VLM: the fine-tuned Vintern-1B baseline of AutoViVQA adds LoRA adapters to the language model and splits every image into up to six high-resolution tiles plus a thumbnail, so that the decoder reads up to 1,792 visual tokens per image. The alternative is to design a task-specific architecture, as ViMoE-VQA [18] does with a sparse mixture-of-experts fusion module and a decoder trained on AutoViVQA. A cheaper option is to keep the whole VLM frozen and train only a small connector between its encoder and decoder, but the design of that connector is delicate. A connector that reads only the global image vector is light, yet it cannot convey small objects, printed text or spatial relations. Connectors that learn to compress the patch tokens, such as Q-Formers, can convey such details but must learn a new alignment to the decoder from limited Vietnamese data.

Vintern-1B already contains a component that solves this alignment problem: its projector, which was pre-trained to map image regions into the input space of the decoder. At the same time, it discards the global [CLS] vector of its encoder, which summarises the whole scene. This observation raises a simple question: *which visual tokens should a frozen Vietnamese VLM receive, and how many?*

To address this question, we propose **ViBridge-VQA**, which gives the frozen decoder of Vintern-1B two complementary kinds of visual tokens from a single image view. *Local* tokens reuse the frozen projector of Vintern-1B, average-pooled to a chosen budget and placed in its original image slot, so the decoder receives region features in the representation it was trained to read. *Global* tokens are produced from the [CLS] vector by a small trainable bridge and add the scene-level information that Vintern-1B ignores. Only this bridge is trained. Because every additional token increases inference cost, we treat the number of global and local tokens as a budget and select it as the knee of the accuracy-cost Pareto front. As a result, ViBridge-VQA with 14 global and 36 local tokens outperforms the officially fine-tuned Vintern-1B on most metrics while reading up to 36 times fewer visual tokens and updating none of its weights.

In summary, our main contributions are as follows:

- We propose **ViBridge-VQA**, a global-local bridge that improves a frozen Vietnamese VLM by combining trainable global tokens with local tokens from

its own pre-trained projector. It raises the F1 of Vintern-1B on AutoViVQA from 17.6 (zero-shot) to 54.6 with 1.35% of the parameters trained, and to 58.2 when combined with a lightweight decoder LoRA.

- We study the **visual token budget** systematically over eight configurations and three random seeds, and select it with an explicit criterion. The knee of the Pareto front falls on the same configuration for every metric, on both evaluation splits, and for cost measured in tokens or in inference time.
- We **analyse where the gains come from**: local tokens matter far more than global ones, should come from the pre-trained projector, complement decoder adaptation, and transfer to four unseen Vietnamese VQA datasets.

2 Related Work

Early VQA systems classified over a fixed set of answers, whereas current generative VLMs produce free-form answers with an autoregressive language model [11,1,17]. For Vietnamese, ViVQA [27] and the open-ended OpenViVQA [20] were followed by datasets for synthetic reasoning (ViCLEVR [29]), book covers (ViOCRQA [24]) and scene text (ViTextVQA [22]). AutoViVQA [19] contributes 37,077 questions on 19,411 COCO images [15], generated by large language models, filtered by an ensemble of validators and annotated with reasoning categories, with an emphasis on relational, causal and multi-step reasoning. On the modelling side, BARTPhoBEiT [28] and AViVQA-TranConI [21] combine pre-trained visual and Vietnamese language representations, Vintern-1B [6] brings an instruction-tuned VLM to Vietnamese, and ViMoE-VQA [18] routes multimodal tokens to sparse experts for generative VQA on AutoViVQA. These approaches either train a new model or fine-tune an existing VLM on multi-tile inputs; in contrast, we keep every pre-trained weight of Vintern-1B frozen and use a single image view.

Keeping the vision encoder and the language model frozen and training only a connector between them is a common way to build VLMs efficiently. Frozen [30] learns a visual prefix for a frozen language model, BLIP-2 [11] trains a Q-Former whose learned queries summarise the image, and LLaVA [17,16] projects every patch token with an MLP. Honeybee [4] argues that a connector should preserve the locality of visual features, and InternVL [5], on which Vintern-1B is built, merges neighbouring patches before its projector. Unlike this work, which trains connectors during large-scale pre-training, we have only a small labelled dataset, and compare connectors that read the global vector or learn to compress the patches with one that reuses the pre-trained projector of the VLM.

The number of visual tokens largely determines the cost of the language decoder. Inference-optimal scaling of VLMs [12] shows that a few tokens can suffice for many tasks, and Matryoshka multimodal models [3] train a single model at several pooled token budgets and show that the required budget depends on the task. Parameter-efficient methods such as adapters [8], prefix tuning [13] and LoRA [9] reduce the cost of adaptation from the other side, by training a small number of additional weights in a frozen network. We bring the two views together: we vary the number of global and local tokens separately, select

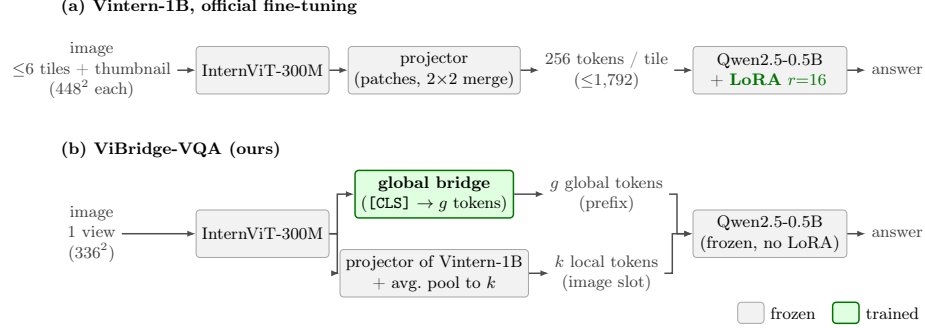


Fig. 1: Overview of ViBridge-VQA compared with the official fine-tuning of Vintern-1B. (a) The official recipe encodes up to seven views per image, passes up to 1,792 visual tokens to the decoder and adapts the decoder with LoRA. (b) ViBridge-VQA encodes a single view and passes g global tokens, produced from the $[\text{CLS}]$ vector by a trainable bridge, and k local tokens, produced by the frozen projector of Vintern-1B and average-pooled. Only the global bridge is trained (1.35% of the parameters for $g = 14$); the selected budget is $g = 14$, $k = 36$.

the budget with explicit criteria, namely the knee of the Pareto front [26] and TOPSIS [10], and ask whether a small training budget is better spent on the visual tokens or on adapting the decoder.

3 Proposed Method

We formulate VQA as conditional answer generation: given an image I and a Vietnamese question q , the model generates a short free-form answer $y = (y_1, \dots, y_T)$. ViBridge-VQA builds on Vintern-1B [6] and keeps all of its pre-trained weights frozen. Instead of adapting the model itself, it changes what the language decoder sees: a set of global tokens produced by a small trainable bridge and a set of local tokens produced by the frozen projector of Vintern-1B.

3.1 Overall Architecture

As illustrated in Fig. 1, ViBridge-VQA keeps the three components of Vintern-1B: the InternViT-300M vision encoder (24 layers, width 1024), its two-layer MLP projector P , and the Qwen2.5-0.5B-Instruct language decoder (24 layers, width 896). Each image is resized to a single 336×336 view, which InternViT splits into 24×24 patches of 14×14 pixels. The encoder returns one 1024-dimensional vector per patch together with a global $[\text{CLS}]$ vector $v \in \mathbb{R}^{1024}$. Vintern-1B uses only the patch vectors: it merges every 2×2 block of neighbouring patches into one 4096-dimensional vector (pixel shuffle) and maps each merged vector into the decoder input space with P , which was pre-trained to align the two backbones. At our resolution this yields a 12×12 grid of 144 region tokens. ViBridge-VQA uses both outputs of the encoder.

3.2 Global Tokens

The [CLS] vector summarises the whole scene, but Vintern-1B never passes it to the decoder. A small trainable bridge g_ϕ expands it into g tokens with two linear heads:

$$g_\phi(v) = [W_a v + b_a; \text{reshape}_{(g-1) \times 896}(W_e v + b_e)] \in \mathbb{R}^{g \times 896}. \quad (1)$$

The first head produces an *anchor* token, and the second produces $g - 1$ complementary tokens that present the global content to the decoder from different learned directions. The global tokens are placed before the prompt. We use $g = 14$ (12.86M parameters): in a preliminary study, the F1 of this bridge alone rose with g up to 14 and then plateaued.

3.3 Local Tokens

Global tokens alone cannot describe small objects, printed text or the position of objects. ViBridge-VQA therefore also passes local tokens, taken directly from the 144 region tokens of the frozen projector P . To control their number, we average-pool the 12×12 grid to an $s \times s$ grid, so that $k = s^2$ with $s \in \{1, 3, 6, 12\}$: $k = 144$ keeps every region token, and $k = 36$ averages each 2×2 block. The pooled tokens replace the embeddings of k image-context placeholders inside Vintern-1B’s own image slot, $\langle \text{img} \rangle \dots \langle / \text{img} \rangle$, in the prompt. Because they come from the pre-trained projector rather than from a newly learned projection of the patches, the local tokens stay in the representation that the decoder was trained to read, and no parameters are needed to produce them.

3.4 Training Objective

Only the bridge parameters ϕ are trained; the vision encoder, the projector and the decoder remain frozen, and no adapter is added to the decoder. We minimise the cross-entropy of the answer tokens:

$$\mathcal{L} = - \sum_{t=1}^T \log p(y_t \mid y_{<t}, g_\phi(v), \text{pool}_s(P(x)), q), \quad (2)$$

where x denotes the merged patch vectors. With $g = 14$, 12.86M parameters (1.35% of the model) are trainable. ViBridge-VQA can also be combined with decoder adaptation: in that variant, a rank-16 LoRA [9] on the query, key, value and output projections of every decoder layer is trained jointly with the bridge, which adds 2.16M parameters (15.02M or 1.57% in total).

3.5 Choosing the Token Budget

The cost of the decoder grows with the number of visual tokens $g + k$, so we treat the pair (g, k) as a budget to be chosen. In our experiments, accuracy

increases monotonically with the number of visual tokens, so every configuration lies on the accuracy–cost Pareto front and the front alone does not single one out. We therefore select its *knee* [26], the point after which additional tokens bring diminishing returns. After normalising cost c and accuracy a to $[0, 1]$ over the front, the knee is the configuration that maximises $\hat{a} - \hat{c}$, i.e., the one lying furthest above the line joining the cheapest and the most accurate configuration. The knee is computed on the validation split only, and we check the choice with the multi-criteria method TOPSIS [10] under several weightings of accuracy and cost.

4 Experiments

We evaluate how much ViBridge-VQA improves Vintern-1B, how many tokens it should use, where its gains come from, and whether they carry over to unseen datasets.

4.1 Dataset and Evaluation Metrics

Dataset. We evaluate on AutoViVQA [19], a Vietnamese VQA benchmark built on MS COCO images [15] with an LLM-driven generation pipeline and ensemble-based quality control. Each question has five short free-form reference answers of 1–10 words and is labelled with its reasoning type. Because the public random 80/20 split lets questions about the same image appear in both training and validation, we remove 273 duplicated (image, question) pairs and 97 questions with irregular type labels, keep the eight remaining reasoning categories, and re-partition the 36,707 questions by image into a 70/15/15 split (seed 42). The resulting training, validation and test sets contain 25,776, 5,463 and 5,468 questions on 13,579, 2,909 and 2,910 images, respectively. This grouped split is slightly harder than a random one (by at most 0.5 F1 in our runs), and all our results are reported on it.

Evaluation metrics. Following AutoViVQA and ViMoE-VQA, we report answer-matching metrics (exact-match Accuracy, and word-level Precision, Recall and F1) and generation metrics (BLEU [23], ROUGE-L [14], METEOR [2] and CIDEr [31]), all multiplied by 100. Word-level scores are computed against each of the five references, keeping the best match. Note that we average F1 over questions, which yields lower values than the harmonic mean of the averaged precision and recall used for the published baselines; computed in the latter way, ViBridge-VQA would obtain 55.95 instead of 54.56 F1. Our F1 comparisons are therefore conservative.

4.2 Experimental Setup

Implementation details. The vision encoder, the projector and the decoder are initialised from Vintern-1B-v3.5 [6] and kept frozen. ViBridge-VQA is trained for

Table 1: Performance comparison on the AutoViVQA test set ($\times 100$). Baselines are taken from AutoViVQA [19] and ViMoE-VQA [18]; ViMoE-VQA is averaged over five seeds. For our models we report the mean $\pm$ standard deviation over three seeds (global only: four), the share of trained parameters, and use 8 (global only), 50 ($k=36$) or 158 ($k=144$) visual tokens, against up to 1,792 for the fine-tuned Vintern-1B. Best results per column are in bold. [†]Far outside the range of all other systems and not used for comparison.

Category	Model	Acc	Prec	Rec	F1	BLEU	ROUGE-L	METEOR	CIDEr
Multimodal (fine-tuned)	ViT5_ViT	7.97	46.84	50.33	48.52	4.13	46.89	31.02	72.68
	BARTPhoBEiT	8.81	45.30	46.48	45.88	43.29 [†]	44.83	24.57	188.96 [†]
	Vintern-1B	13.01	52.47	55.12	53.76	6.11	51.93	35.25	72.84
	ViMoE-VQA	9.55	62.89	58.60	60.69	12.54	47.12	39.10	88.76
LLMs and VLMs (zero-shot)	Vintern-1B	0.12	17.52	19.87	17.55	1.91	25.84	23.93	8.54
	LLaMA 3.2	0.36	23.96	73.71	36.16	3.62	36.11	30.01	62.84
	Gemini 2.0 Flash	0.55	27.20	74.10	39.79	4.41	39.60	31.72	74.42
	Gemini 2.5 Flash	0.22	24.43	76.66	24.75	0.39	37.27	31.22	71.90
	GPT-5	10.84	47.20	55.20	50.89	6.07	47.30	33.34	84.20
Global only (ours, $k=0$)	$g=8$ (0.78%)	8.86 $\pm$.37	51.19 $\pm$.19	52.37 $\pm$.21	50.47 $\pm$.22	16.24 $\pm$.43	48.70 $\pm$.19	41.10 $\pm$.29	96.19 $\pm$ 1.10
	+ LoRA, 3 ep. (1.01%)	11.40 $\pm$.24	55.37 $\pm$.18	56.16 $\pm$.09	54.52 $\pm$.14	20.97 $\pm$.29	52.68 $\pm$.21	45.08 $\pm$.14	108.43 $\pm$.67
ViBridge-VQA (ours, $g=14$)	$k=36$ (1.35%)	11.75 $\pm$.21	55.36 $\pm$.21	56.56 $\pm$.21	54.56 $\pm$.19	18.52 $\pm$.18	52.64 $\pm$.19	45.53 $\pm$.33	110.58 $\pm$.95
	$k=144$ (1.35%)	13.19 $\pm$.23	56.64 $\pm$.13	58.17 $\pm$.26	56.04 $\pm$.19	19.79 $\pm$.15	54.11 $\pm$.16	47.38 $\pm$.26	116.74 $\pm$.76
	$k=36$ + LoRA, 1 ep. (1.57%)	14.67 $\pm$.30	59.02 $\pm$.12	59.96 $\pm$.33	58.17 $\pm$.19	22.53 $\pm$.11	56.25 $\pm$.22	49.32 $\pm$.37	122.88 $\pm$.93

two epochs, after which neither the validation loss nor CIDEr improves further; the variants with a decoder LoRA are trained for one or three epochs. We use AdamW with a learning rate of 2×10^{-4} and an effective batch size of 8, train on the first reference answer, and decode greedily. Unless stated otherwise, every configuration is trained with three random seeds (42, 123 and 3407), and we report the mean and the standard deviation. All runs fit on a single 16 GB GPU (Tesla P100 or T4). The token budget is selected on the validation split, and the test split is used only for the final report.

Baselines. We compare against three groups of systems: (1) multimodal models fine-tuned on AutoViVQA, namely ViT5_ViT, BARTPhoBEiT [28], Vintern-1B and ViMoE-VQA [18]; (2) models used zero-shot, namely Vintern-1B, LLaMA 3.2, Gemini 2.0 and 2.5 Flash, and GPT-5; and (3) our *global-only* model, which uses the same bridge without local tokens ($k = 0$, $g = 8$), with and without a decoder LoRA. The results of groups (1) and (2) are taken from the AutoViVQA and ViMoE-VQA papers and were obtained on an 8:1:1 split of the original release. The fine-tuned Vintern-1B follows the official recipe: the vision encoder and the projector are frozen, a rank-16 LoRA adapts the language model for one epoch, and each image is encoded as up to six 448×448 tiles plus a thumbnail.

4.3 Main Results

Table 1 compares ViBridge-VQA with the baselines. Without updating any weight of Vintern-1B, ViBridge-VQA raises its F1 from 17.55 in the zero-shot setting to 54.56, and its CIDEr from 8.54 to 110.58. With the selected budget of 50 visual tokens, it already outperforms the officially fine-tuned Vintern-1B on seven of the eight metrics; the largest gains appear in the generation metrics (BLEU,

Table 2: Effect of the token budget ($\times 100$, mean over three seeds, $\pm$ standard deviation for F1; global only: four seeds). Tokens: visual tokens passed to the decoder ($g + k$). Time: minutes for one evaluation session on the 10,931 validation and test questions, including model loading (Kaggle P100/T4). The selected budget is in bold.

g	k	Tokens	Time	Validation		Test	
				F1	CIDEr	F1	CIDEr
8	0	8	–	50.74 $\pm$.17	99.07	50.47 $\pm$.22	96.19
8	1	9	57.0	51.03 $\pm$.12	99.27	50.71 $\pm$.16	96.90
8	9	17	59.2	52.82 $\pm$.11	106.11	52.11 $\pm$.25	101.30
8	36	44	60.6	54.32 $\pm$.41	111.88	53.82 $\pm$.22	107.67
8	144	152	71.2	56.15 $\pm$.08	119.05	55.20 $\pm$.07	114.13
14	1	15	58.8	51.67 $\pm$.25	101.66	51.67 $\pm$.12	99.53
14	9	23	57.7	52.98 $\pm$.08	106.02	52.76 $\pm$.28	103.63
14	36	50	60.0	54.96 $\pm$.28	113.29	54.56 $\pm$.19	110.58
14	144	158	70.2	56.61 $\pm$.04	120.71	56.04 $\pm$.19	116.74

METEOR and CIDEr), and only exact-match accuracy remains 1.3 points lower. When all 144 local tokens are used, the model surpasses the fine-tuned Vintern-1B on every metric, including accuracy, while still reading up to eleven times fewer visual tokens. These results suggest that much of what AutoViVQA requires is already present in the frozen Vintern-1B, provided that its decoder receives suitable visual input.

Decoder adaptation adds to these gains rather than replacing them. Training ViBridge-VQA jointly with a rank-16 LoRA for a single epoch raises test F1 by a further 3.6 points to 58.17 and gives the best accuracy, BLEU, ROUGE-L, METEOR and CIDEr among all compared systems (setting aside the outlying BARTPhoBEiT values). The task-specific ViMoE-VQA, whose fusion module and decoder are trained on AutoViVQA, keeps the highest precision and F1, 2.5 F1 points above our best model. Zero-shot language models reach a high recall with long answers but lose precision, whereas our answers match the reference length (Section 4.6).

Our global-only model isolates the effect of the local tokens: even with a frozen decoder, ViBridge-VQA matches the global-only model trained with a three-epoch decoder LoRA in F1 (54.56 versus 54.52) and exceeds it in CIDEr (110.58 versus 108.43), although its BLEU is lower.

Statistical stability. All our models behave consistently across random seeds. On the test split, the standard deviation of ViBridge-VQA is at most 0.19 for F1 and 0.95 for CIDEr, and for each of our models in Table 1, test F1 lies at most 0.6 points below validation F1, which indicates that the budget selected on the validation split does not over-fit it.

4.4 Choosing the Token Budget

Table 2 and Fig. 2a vary the number of global tokens $g \in \{8, 14\}$ and of local tokens $k \in \{1, 9, 36, 144\}$. The most striking pattern is that *local tokens matter*

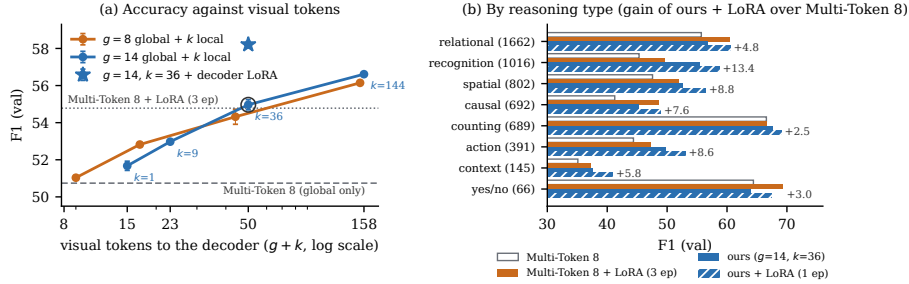


Fig. 2: (a) Validation F1 against the number of visual tokens (mean $\pm$ std over three seeds); the circled point is the knee, $g = 14$, $k = 36$. Horizontal lines: the global-only model (four seeds) and the same model with a three-epoch decoder LoRA (three seeds). (b) Validation F1 by reasoning type (number of questions in parentheses): global only and + LoRA for seed 42, ViBridge-VQA ($g = 14$, $k = 36$) without and with a one-epoch decoder LoRA averaged over three seeds; the numbers give the gain of ViBridge-VQA + LoRA over the global-only model.

far more than global ones. At $g = 14$, increasing k from 1 to 144 adds 4.9 F1 and 19.1 CIDEr on the validation split, whereas increasing g from 8 to 14 at a fixed k adds only 0.2–0.6 F1. Accuracy has not saturated within the 144 region tokens available at our resolution, as each step of k still adds 1.3–2.0 F1, but the gain per token drops quickly, from 0.07 F1 per token between 23 and 50 tokens to 0.015 between 50 and 158. The budget is therefore a genuine trade-off between accuracy and cost.

Because accuracy grows with every added token, all configurations lie on the Pareto front, and we select its knee (Section 3.5). On the validation split, the knee is $g = 14$, $k = 36$. Remarkably, the same configuration is the knee for each of the eight metrics on both validation and test, whether cost is counted in visual tokens or measured as inference time; with a logarithmic token axis, it remains the knee in 11 of the 16 cases. TOPSIS leads to the same conclusion when accuracy and cost are weighted equally, using the gains in F1, CIDEr and METEOR as benefits and the added tokens, inference time, trained parameters and seed variance as costs. Weighting cost more favours $k = 9$, and weighting accuracy more favours $k = 144$, but $g = 14$, $k = 36$ is the only configuration that stays among the top four for every accuracy weight between 0.3 and 0.7. In practical terms, it obtains 72% of the F1 gain of $k = 144$ over the global-only model with 32% of its visual tokens, and it is the cheapest configuration that reaches the global-only model with a three-epoch decoder LoRA. Fourteen global tokens are better than eight at every k , so we use $g = 14$.

Computational efficiency. ViBridge-VQA encodes a single 336×336 view per image, which costs 362 GFLOPs and 229 ms with InternViT-300M on a Tesla P100. The six 448×448 tiles and the thumbnail used by the fine-tuned Vintern-1B cost 2,172 GFLOPs and 922 ms, and give the decoder up to 1,792 visual tokens

Table 3: Where the visual tokens come from, with and without decoder adaptation (validation, $\times 100$, mean over three seeds with $\pm$ std for F1; Multi-Token: four seeds; *seed 42 only). Bridge-only models are trained for two epochs; + LoRA is a rank-16 decoder LoRA trained jointly with the bridge for one epoch (2.16M additional parameters). Params: trained bridge parameters.

Visual tokens from	Tokens	Params	Bridge only		+ decoder LoRA		
			F1	CIDEr	F1	CIDEr	
<i>Global vector ([CLS])</i>							
Residual	1	4.86M	46.02 $\pm$.36	86.48	52.69 $\pm$.14	104.25	
Multi-Token	8	7.35M	50.74 $\pm$.17	99.07	53.47 $\pm$.17	106.68	
<i>Learned compression of the patch tokens</i>							
Tile-Attention	8	4.14M	45.14 $\pm$.92	84.14	52.82*	104.72*	
Light Q-Former	8	27.6M	47.06 $\pm$.66	88.07	53.21 $\pm$.14	106.26	
Full Q-Former	16	69.4M	46.84 $\pm$.47	87.74	53.29 $\pm$.29	106.50	
<i>Global vector + pre-trained projector (ViBridge-VQA, g = 14)</i>							
k = 36	50	12.86M	54.96 $\pm$.28	113.29	58.21 $\pm$.16	124.88	
k = 144	158	12.86M	56.61 $\pm$.04	120.71	—	—	

instead of 50. Keeping all 144 local tokens instead of 36 makes evaluation about 17% slower.

4.5 Ablation Study

Where should the visual tokens come from? Table 3 compares three sources of visual tokens for the frozen decoder. The first group reads only the global [CLS] vector, through a *Residual* bridge that produces one token or the *Multi-Token* bridge of Eq. (1). The second group learns to compress the patch tokens, either with a *Tile-Attention* bridge in which eight learned queries attend over the patches, or with a light or full Q-Former [11]. Without decoder adaptation, the Multi-Token bridge is the best of these five, and the bridges that compress the patches are 3.7–5.6 F1 lower despite having up to $9.4\times$ more parameters. Learning a new alignment of the patch tokens from 25k questions thus does not pay off. ViBridge-VQA, which takes the local tokens from the pre-trained projector instead, improves over the best global-only bridge by 4.2 F1 with $k = 36$ and by 5.9 F1 with $k = 144$. The same pooling operation therefore helps when it is applied to the output of the pre-trained projector, whereas in a preliminary study, pooling the raw patch tokens through a newly learned projection lost 5.1–13.9 F1 and a Honeybee-style convolutional abstractor [4] remained 0.7 F1 below the Multi-Token bridge with $2.7\times$ its parameters.

Local tokens and decoder adaptation are complementary. A decoder LoRA improves every global-only bridge and brings them within 0.8 F1 of one another (52.69–53.47), which suggests that the decoder learns to compensate for poor visual input. Even after this adaptation, all of them remain below ViBridge-VQA with a frozen decoder. Decoder adaptation is therefore not needed to surpass them, but it is still useful: in the same one-epoch joint training, it adds 3.3 F1 to ViBridge-VQA, more than the 2.7 F1 it adds to the Multi-Token bridge.

Conversely, distilling the projector into the global-only bridge, at the feature or output level, lowers F1 by 0.3–0.4 points: the projector’s information is best passed as tokens.

4.6 Analysis by Reasoning Type and Error Analysis

Reasoning types. Fig. 2b breaks the results down by the reasoning categories of AutoViVQA. Local tokens help most where the answer depends on what is visible in the image: recognition (+10.2 F1 over the global-only model), action (+5.4) and spatial (+5.1) questions. On these categories, ViBridge-VQA with a frozen decoder is also better than the global-only model with a decoder LoRA. Conversely, decoder adaptation is more useful for causal and relational questions, whose reference answers are the longest (5.0 and 5.6 words on average), and for yes/no questions. Adding the LoRA to ViBridge-VQA combines both strengths: the resulting model is the best in seven of the eight categories, matching the global-only model with LoRA on causal and relational questions while keeping the large gain of the local tokens on recognition (58.7 versus 49.7 F1). Local tokens improve what the model perceives, whereas decoder adaptation improves how it formulates reasoning answers.

Error analysis. An analysis of our validation predictions helps explain the low exact-match accuracy of all systems. Our answers are as long as the references on average (4.3 words), so the gains do not come from changing the answer length. Most predictions (77–81%) share some, but not all, words with the closest reference: a correct answer is often phrased differently from all five references. Local tokens turn some of these partial answers into exact ones and reduce complete failures: compared with the global-only model, ViBridge-VQA increases the share of predictions with $F1 = 1$ from 9.8% to 13.3% and reduces the share with no word in common with any reference from 9.6% to 8.0% (6.8% with LoRA). The remaining failures concentrate on context, recognition, spatial and action questions (12–18% without any overlap), i.e., on fine-grained perception, whereas they are rare for counting, causal and relational questions ($< 5\%$), where a partially correct answer is easier to produce.

4.7 Generalisation to Other Datasets

Table 4 evaluates the models without further training on four Vietnamese VQA datasets: two scene-text datasets, ViTextVQA [22] and OpenViVQA [20], and two general-domain ones, ViVQA-X [7] and ViVQA [27]. We include the strongest global-only model, trained with a three-epoch decoder LoRA, as a reference. Local tokens change the picture most on scene text, where the global-only model collapses, presumably because the [CLS] vector cannot convey printed words. On ViTextVQA, F1 rises from 3.96 to 15.14 with 36 local tokens and to 33.26 with 144, slightly above the zero-shot Vintern-1B that reads six 448×448 tiles, although ViBridge-VQA reads a single 336×336 view. On OpenViVQA, whose

Table 4: Zero-shot transfer to Vietnamese VQA datasets not seen during training ($\times 100$; about 1,000 sampled test questions per sampling seed, 505–519 for ViVQA; mean over three sampling seeds, $\pm$ std for F1). Vintern-1B reads six tiles and a thumbnail; the other models are trained on AutoViVQA only (seed 42). Best results per column are in bold.

Model	ViTextVQA		OpenViVQA		ViVQA-X		ViVQA	
	F1	CIDEr	F1	CIDEr	F1	CIDEr	F1	CIDEr
Vintern-1B, zero-shot	32.01 $\pm$.31	195.62	58.40 $\pm$.80	411.63	14.52 $\pm$.47	50.13	23.90 $\pm$.24	78.63
Global only + LoRA, 3 ep.	3.96 $\pm$.09	15.64	21.56 $\pm$.48	83.21	15.89 $\pm$.65	42.12	31.48 $\pm$ 1.15	84.94
ViBridge-VQA, $k=36$	15.14 $\pm$.47	66.39	28.24 $\pm$.79	128.36	24.91 $\pm$.14	79.51	42.68 $\pm$ 1.52	171.21
ViBridge-VQA, $k=144$	33.26 $\pm$.67	190.30	35.89 $\pm$.80	187.00	27.89 $\pm$.13	92.22	44.95 $\pm$.90	191.28
ViBridge-VQA, $k=36$ + LoRA	15.39 $\pm$.39	70.07	28.99 $\pm$.51	130.46	27.64 $\pm$.11	95.29	44.76 $\pm$ 1.21	183.60

references are long descriptive sentences, the local tokens narrow the gap to Vintern-1B, but Vintern-1B remains clearly better.

On the general-domain datasets, ViBridge-VQA outperforms both reference models, by 10.4 (ViVQA-X) and 18.8 (ViVQA) F1 over the zero-shot Vintern-1B with $k = 36$. As on AutoViVQA, the zero-shot model reaches a high recall with long answers (61.2 on ViVQA-X) but a very low precision (9.1). Adding the decoder LoRA to ViBridge-VQA helps on these general-domain datasets, bringing $k = 36$ close to $k = 144$, but barely on scene text. Reading text thus requires more local tokens, which decoder adaptation cannot replace, consistent with the token-budget results above.

4.8 Qualitative Analysis

In Examples 1–3 of Fig. 3, only ViBridge-VQA answers correctly, and each answer depends on a local detail: the brand printed on the laptop, the canal that identifies Venice, and the water sprayed by the elephant. The global-only models instead answer “Apple”, “Hanoi” and “swimming”, which are plausible for the scene as a whole but wrong for the image. Example 4 shows a typical failure: ViBridge-VQA describes the scene instead of explaining the purpose of the bus, which both global-only models infer. This matches the advantage of decoder adaptation on causal questions.

5 Conclusion

In this work, we introduced **ViBridge-VQA**, a parameter-efficient approach that improves the Vietnamese VLM Vintern-1B for generative visual question answering without updating any of its pre-trained weights. Instead of adapting the model on many high-resolution tiles, ViBridge-VQA changes what its frozen decoder sees: trainable global tokens from the [CLS] vector and local tokens from the pre-trained projector. By selecting their number as the knee of the accuracy–cost Pareto front, we obtained a single configuration, 14 global and 36 local tokens, that is the knee for every metric and both evaluation splits. Experiments on AutoViVQA show that this configuration outperforms the officially fine-tuned

(1)	(2)	(3)	(4)
			
Q (VI): Chiếc máy tính xách tay này của hãng nào?	Q (VI): Chiếc thuyền đang đi qua thành phố nào?	Q (VI): Con voi đang làm hành động gì với chiếc vòi của nó?	Q (VI): Xe buýt dừng lại để làm gì?
<i>Q (EN): Which brand is this laptop?</i>	<i>Q (EN): Which city is the boat passing through?</i>	<i>Q (EN): What is the elephant doing with its trunk?</i>	<i>Q (EN): Why has the bus stopped?</i>
GT: Dell	GT: Thành phố Venice (Venice)	GT: Phun nước (spraying water)	GT: Để đón hành khách (to pick up passengers)
Global: Apple	Global: Hà Nội (Hanoi)	Global: Đang bơi (swimming)	Global: Đi đón khách (going to pick up passengers)
+LoRA: Apple	+LoRA: Thành phố New Orleans (New Orleans)	+LoRA: Đang dùng vòi (using its trunk)	+LoRA: Để đón khách (to pick up passengers)
Ours: Dell	Ours: Thành phố Venice (Venice)	Ours: Đang phun nước (spraying water)	Ours: Đi dọc đường (going along the road)

Fig. 3: Qualitative examples from the AutoViVQA validation set (seed 42). Q: question, GT: one of the five reference answers, Global: global-only model, +LoRA: the same model with a three-epoch decoder LoRA, Ours: ViBridge-VQA ($g = 14$, $k = 36$). Plausible answers are shown in green and wrong ones in red.

Vintern-1B on seven of eight metrics while reading up to 36 times fewer visual tokens, and that combining it with a lightweight decoder LoRA yields the highest accuracy, ROUGE-L and METEOR among the compared systems. Our analyses show that local tokens matter far more than global ones and complement decoder adaptation.

Our study also has limitations. ViMoE-VQA, a model trained specifically for AutoViVQA, still achieves a higher F1, and the published baselines were evaluated on a different split of AutoViVQA than ours. Out of distribution, the local tokens allow a single 336×336 view to match the six-tile Vintern-1B on the scene-text ViTextVQA, but Vintern-1B remains clearly better on OpenViVQA. Finally, accuracy had not saturated at the 144 region tokens available at our resolution; higher-resolution inputs, which we plan to explore, could bring further gains, especially for reading text.

Overall, our findings show that an existing Vietnamese VLM can be improved substantially by choosing carefully which visual tokens its frozen decoder receives, and that the number of these tokens can be set with an explicit accuracy–cost criterion. The code and the grouped split are available at <https://github.com/nguyennn263/modular-vlm-finetune>. We hope that this work will encourage efficient and carefully evaluated multimodal models for low-resource languages.

Acknowledgments. This research is funded by University of Science, VNU-HCM under grant number CNTT 2025-02.

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